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Waruni Kekulandara

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Research Interests

- Human-Computer Interaction
- Explainable and Interpretable AI
- Human-Centred Artificial Intelligence
- Trust in Intelligent Systems.

Education

University of Peradeniya, B.Sc. Honours in Computer Science 2021 – 2025
Minor in Chemistry and Biology

- GPA: 3.80 / 4.00 (Top of the Batch | First-Class Honours)
- **Relevant Coursework:** Object Oriented Analysis and Design, Software Engineering, Computer Graphics, Artificial Intelligence

Publications

Peer-Reviewed Conference Papers

- **Kekulandara, W.**, Amarasinghe, K., & Dayaratna, E. (2026). *Framework for Existing Counterfactual Methods with Feasibility and LLM Explanation Support.* In ***Proceedings of the 11th International Congress on Information and Communication Technology (ICICT 2026)***, DOI: 10.1007/978-3-032-35982-7_20.
- **Kekulandara, W.**, Amarasinghe, K., & Dayaratna, E. (2025). *Towards Counterfactual Verification for Tree Ensembles.* **2025 IEEE International Conference on Emerging Trends in Engineering and Computing (ETECOM).** DOI: 10.1109/ETECOM66111.2025.11318990
- **Kekulandara, W.**, De Silva, C., & Dayaratna, E. (2026). *Shades of Sentiment: Evaluating Emotional Fairness Across Ethnic Identities in LLMs.* **World Conference on Information Systems for Business Management (ISBM)**, Bangkok, Thailand. *(Accepted)*
- De Silva, C., **Kekulandara, W.**, & Dayaratna, E. (2026). *Is More Explanation Better? A Multi-Dimensional Study of Combined XAI Techniques for Regression.* **8th IEEE International Conference on Advancements in Computing 2026**, Colombo, Sri Lanka. *(In Review)*

Conference Abstracts

- **Kekulandara, W.**, Amarasinghe, K., & Dayaratna, E. (2025). *A Unified Framework for Evaluating Existing Counterfactual Methods with Feasibility Constraints and LLM-Based Explanations.* **Sri Lanka Japan Collabarative Research (SLJCR 2025).** *(Presented)*

Awards & Honors

Alumni Prize for Excellence in Computer Science, University of Peradeniya 2026

- Awarded for achieving the highest GPA and graduating top of the Special Degree in Computer Science with an 'A' in the Research course.

University Award for Academic Excellence, University of Peradeniya 2026

Research Experience

Final Year Research Project, University of Peradeniya 2025 – 2026
Enhancing Counterfactual Extraction Methods for Tree Ensembles

- Designed and implemented an evaluation framework for counterfactual explanation methods.
- Evaluated five selected counterfactual methods (Foil Trees, SHAP-C, LIME-C, LORE, CLEAR).
- Implemented an interactive Python UI incorporating Llama for visualizing counterfactual results.

- This project is the basis of three conference publications/presentations (ETECOM 2025, SLJCR 2025, ICICT 2026).

EEG Correlations of Creativity, Personality, and Vocational Interest: A Scoping Review, University of Peradeniya — Ongoing

- Conducting an EEG experiment with university students to build a novel EEG dataset.
- Co-authoring an accompanying review paper as part of the same project.

LLM Agent for Emotional Fairness Across Ethnic Identities, University of Peradeniya — Ongoing

- Extending the *Shades of Sentiment* study (accepted, ISBM 2026) by developing an autonomous agent framework to evaluate and mitigate emotional fairness across ethnic identities in LLMs.

Ethical and Technological Governance Framework Against coordinated harassment of celebrities in Digital Ecosystems, University of Peradeniya — Ongoing

- Investigating platform moderation and policy approaches to online harassment targeting K-pop idols and other online public figures.

Physics-Informed Machine Learning for fMRI Hemodynamic Modeling, University of Peradeniya — Ongoing

- Developing a physics-informed machine learning model to estimate Balloon-model parameters from auditory fMRI data, integrating physiological modeling with data-driven learning.

Teaching Experience

Graduate Teaching Assistant, Faculty of Science, University of Peradeniya — Jan 2026 – Present

- Conducted practical laboratory sessions in Programming Laboratory II (Java), Computing for Science (Python), and Computer Graphics Programming (C++/OpenGL).
- Delivered lectures and facilitated academic discussions for theory courses, including Computer Architecture, Introduction to Computer Networks, Object-Oriented Programming, and Digital Image Processing.
- Developed practical assignments and examinations and assessed coursework and final examinations for a cohort of 250+ students.

Workshop Instructor, Postgraduate Institute of Science, University of Peradeniya

- Delivered a workshop for Master's students on explainable AI, covering SHAP, LIME, and counterfactual explanation methods.

Workshop Facilitator, Kingswood College, Kandy

- Facilitated a hands-on workshop introducing IoT concepts to school students.

Professional Experience

Project Manager, NoLoopTech (Pvt) Ltd — Jun 2025 – Jul 2026

- Managed software products from planning through deployment.
- Acted as the primary point of contact between clients, stakeholders, and internal teams.
- Led requirement gathering and sprint planning for AI and web application projects.

Intern, Innovator Park International (Pvt) Ltd — Oct 2023 – Feb 2024

- Taught basic Science (Biology, Chemistry) and introductory IoT concepts to primary and middle school students.

Selected Additional Projects

- **Health-Mate:** Full-stack web application designed and built end-to-end (React, TypeScript, Java, PostgreSQL).
- **Area Calculation and Division in Survey Plans:** Automated land-area computation from survey-plan images using Python, OpenCV, NumPy, and Matplotlib.
- **Effect of Light and Temperature on Dye Desegregation:** Experimental study of dye stability in food, hair, and denim colorings using UV/Visible spectroscopy.
- **eBook Hosting Website:** PHP-based platform for hosting student-authored eBooks, with front-end and database design.
- **Sydney Opera House 3D Model:** Detailed 3D reconstruction using OpenGL, C++, GLUT, and SOIL.

Technical Skills

- **Programming Languages:** Python, Java, C, C++, JavaScript
- **AI/ML:** NumPy, Pandas, Scikit-learn, Matplotlib, PyTorch, TensorFlow, SHAP, LIME
- **Databases:** MySQL, PostgreSQL
- **Tools:** Git/GitHub, Visual Studio Code, IntelliJ, Google Colab
- **Frameworks & Web:** React, Spring Boot, RESTful APIs, HTML, CSS

Leadership & Service

- **Editor**, Astronomical Society, University of Peradeniya (2025 – 2026)
- **Program Team Member**, IEEE Computer Society, University of Peradeniya (2024 – 2025)
- **Junior Treasurer**, Drama Society, University of Peradeniya (2023 – 2024)
- **President Guide**, Sri Lanka Girl Guides Association (2013 – 2019)

Volunteering

- RESCON, Postgraduate Institute of Science, University of Peradeniya (2026)
- Research Cafe, University of Peradeniya (2026)
- Huddle Workshop, University of Peradeniya (2026, 2024, 2023, 2022)
- Technical Support for E-Book Project, Mahamaya Girls' College, Kandy (2025)
- Teaching Science & Mathematics to O/L Students, Uduwela Buddhist College (2023)
- Project for Special Needs Children, Blue Rose School (2023)
- Save the Next Generation - Knuckles, University of Peradeniya (2022)

Languages

English (IELTS:8) | Sinhala (Native) | Tamil (Reading) | Korean (Reading)

References

Dr. Erunika Dayaratna
Senior Lecturer in Computer Science,
Department of Statistics and Computer Science
University of Peradeniya
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